



Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
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Laporan Akhir

Praktikum Jaringan Komputer

Konfigurasi Vlan, OSPF Multivendor

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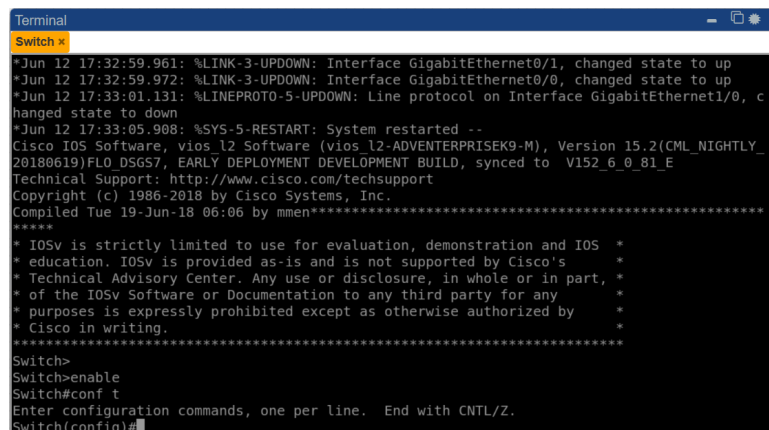
2026

1 Langkah Percobaan

Pada praktikum ini dilakukan konfigurasi routing dinamis OSPF pada jaringan multivendor yang terdiri dari Cisco Router, Huawei Router, dan Mikrotik Router. Konfigurasi dilakukan untuk membangun pertukaran informasi routing secara otomatis serta menyediakan jalur redundan antarjaringan. Setelah konfigurasi selesai, dilakukan pengujian konektivitas dan failover dengan mensimulasikan gangguan pada link utama untuk memastikan OSPF dapat melakukan konvergensi dan mengalihkan trafik ke jalur cadangan secara otomatis.

1.1 Konfigurasi Cisco Switch

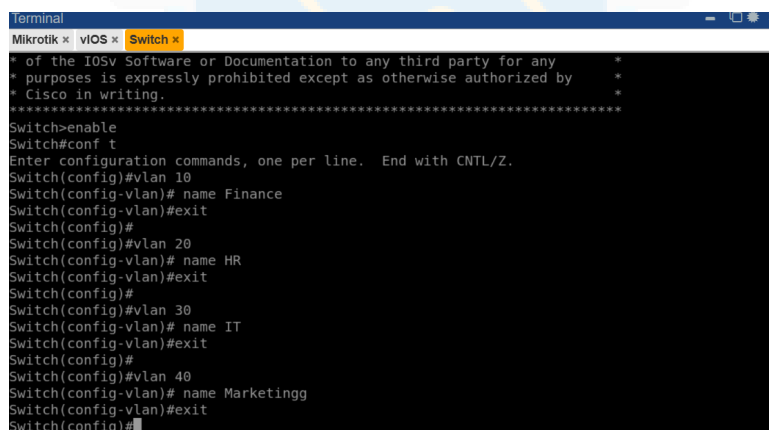
1. Masuk ke terminal Cisco Switch



```
Terminal
Switch >
*Jun 12 17:32:59.961: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
*Jun 12 17:32:59.972: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Jun 12 17:33:01.131: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0, c
changed state to down
*Jun 12 17:33:05.908: %SYS-5-RESTART: System restarted --
Cisco IOS Software, vios l2 Software (vios l2-ADVENTERPRISEK9-M), Version 15.2(CML_NIGHTLY_
20180619)FLO_DSG57, EARLY DEPLOYMENT DEVELOPMENT BUILD, synced to V152_6_0_81_E
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2018 by Cisco Systems, Inc.
Compiled Tue 19-Jun-18 06:06 by mmen*****
*****
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* Technical Advisory Center. Any use or disclosure, in whole or in part, *
* of the IOSv Software or Documentation to any third party for any *
* purposes is expressly prohibited except as otherwise authorized by *
* Cisco in writing. *
*****
Switch>
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
```

Gambar 1: Login Ke terminal Cisco

2. Membuat Vlan



```
Terminal
Mikrotik x viOS x Switch >
* of the IOSv Software or Documentation to any third party for any *
* purposes is expressly prohibited except as otherwise authorized by *
* Cisco in writing. *
*****
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)# name Finance
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 20
Switch(config-vlan)# name HR
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 30
Switch(config-vlan)# name IT
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 40
Switch(config-vlan)# name Marketingg
Switch(config-vlan)#exit
Switch(config)#
```

Gambar 2: Konfigurasi vlan cisco

3. Pastikan vlan sudah dibuat dan statusnya aktif

```

Terminal
Mikrotik x vIOS x Switch x
Switch(config-vlan)# name Marketingg
Switch(config-vlan)#exit
Switch(config)#exit
Switch#show c
*Jun 12 17:24:51.979: %SYS-5-CONFIG_I: Configured from console by consolelan
^
% Invalid input detected at '^' marker.

Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Gi0/0, Gi0/1, Gi0/2, Gi0/3
                                           Gi1/0, Gi1/1, Gi1/2, Gi1/3
10   Finance                active
20   HR                    active
30   IT                    active
40   Marketingg            active
1002 fddi-default          act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default          act/unsup
Switch#

```

Gambar 3: vlan status aktif

4. Konfigurasi Access port untuk setiap vlan

```

Terminal
Switch x
% Invalid input detected at '^' marker.

Switch(config)#
Switch(config)#interface gi0/2
Switch(config-if)# description PC-FINANCE-VLAN10
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface gi0/1
Switch(config-if)# description PC-HR-VLAN20
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface gi1/1
Switch(config-if)# description PC-IT-VLAN30
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface gi0/3
Switch(config-if)# description PC-MARKETING-VLAN40
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 40
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#

```

Gambar 4: Konfigurasi Access port

5. Konfigurasi Trunk ke Cisco Router

```

Terminal
Switch x
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface gi0/0
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# description TRUNK-T0-CISCO-ROUTER
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#interface gi0/0
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# description TRUNK-T0-CISCO-ROUTER
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 20
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#interface gi0/0
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# description TRUNK-T0-CISCO-ROUTER
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 30
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#interface gi0/0
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# description TRUNK-T0-CISCO-ROUTER
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 40
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#

```

Gambar 5: Konfigurasi Trunk di router

```
Terminal
Switch >
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Gi1/0, Gi1/2, Gi1/3
10   Finance                 active    Gi0/2
20   HR                     active    Gi0/1
30   IT                     active    Gi1/1
40   Marketingg             active    Gi0/3
1002 fddi-default          act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default          act/unsup
Switch#show interfaces trunk

Port      Mode          Encapsulation  Status        Native vlan
Gi0/0     on            802.1q         trunking      1

Port      Vlans allowed on trunk
Gi0/0     10,20,30,40

Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10,20,30,40
Switch#
```

Gambar 6: Hasil Show Trunk

1.2 Konfigurasi Cisco Router

1. Masuk ke Cisco Router

```
Terminal
vIOS >
*****
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* education. IOSv is provided as-is and is not supported by Cisco's *
* Technical Advisory Center. Any use or disclosure, in whole or in part, *
* of the IOSv Software or Documentation to any third party for any *
* purposes is expressly prohibited except as otherwise authorized by *
* Cisco in writing. *
*****
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
```

Gambar 7: Login ke cisco router

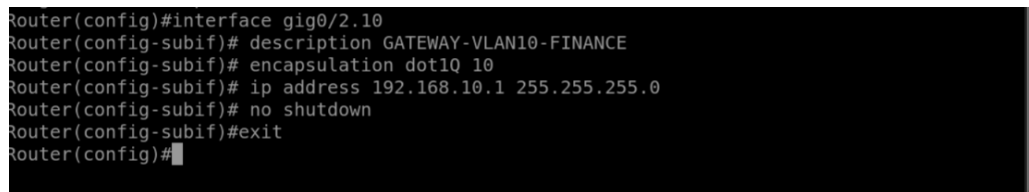
2. Konfigurasi Interface Trunk ke Switch

A terminal window titled 'Terminal' with a 'vIOS x' tab. It displays the configuration of a trunk interface on a Cisco switch. The output shows a disclaimer about IOSv being for evaluation purposes, followed by the configuration commands: 'enable', 'conf t', 'interface gig0/2', 'description TRUNK-T0-CISCO-SWITCH', 'no ip address', 'no shutdown', and 'exit'.

```
*****
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* education. IOSv is provided as-is and is not supported by Cisco's *
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* of the IOSv Software or Documentation to any third party for any *
* purposes is expressly prohibited except as otherwise authorized by *
* Cisco in writing. *
*****
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/2
Router(config-if)# description TRUNK-T0-CISCO-SWITCH
Router(config-if)# no ip address
Router(config-if)# no shutdown
Router(config-if)# exit
```

Gambar 8: Konfigurasi Trunk di switch

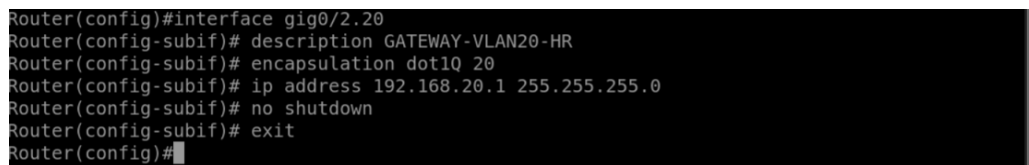
3. Konfigurasi Subinterface VLAN 10

A terminal window showing the configuration of a subinterface for VLAN 10. The commands are: 'interface gig0/2.10', 'description GATEWAY-VLAN10-FINANCE', 'encapsulation dot1Q 10', 'ip address 192.168.10.1 255.255.255.0', 'no shutdown', 'exit', and 'Router(config)#'.

```
Router(config)#interface gig0/2.10
Router(config-subif)# description GATEWAY-VLAN10-FINANCE
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)# exit
Router(config)#
```

Gambar 9: Konfigurasi Subinterface

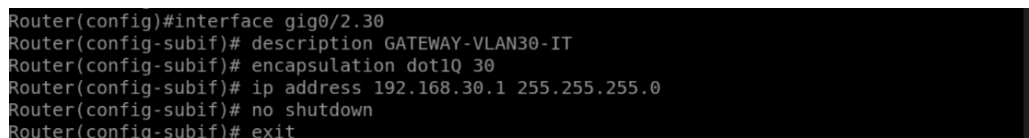
4. Konfigurasi Subinterface VLAN 20

A terminal window showing the configuration of a subinterface for VLAN 20. The commands are: 'interface gig0/2.20', 'description GATEWAY-VLAN20-HR', 'encapsulation dot1Q 20', 'ip address 192.168.20.1 255.255.255.0', 'no shutdown', 'exit', and 'Router(config)#'.

```
Router(config)#interface gig0/2.20
Router(config-subif)# description GATEWAY-VLAN20-HR
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)# exit
Router(config)#
```

Gambar 10: Konfigurasi Subinterface

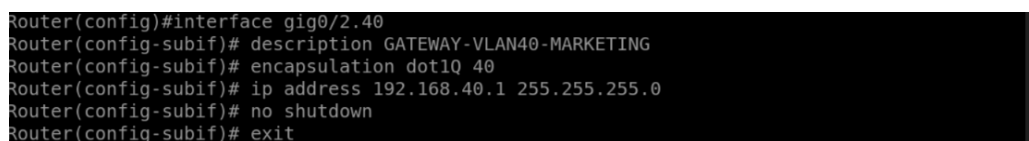
5. Konfigurasi Subinterface VLAN 30

A terminal window showing the configuration of a subinterface for VLAN 30. The commands are: 'interface gig0/2.30', 'description GATEWAY-VLAN30-IT', 'encapsulation dot1Q 30', 'ip address 192.168.30.1 255.255.255.0', 'no shutdown', 'exit', and 'Router(config)#'.

```
Router(config)#interface gig0/2.30
Router(config-subif)# description GATEWAY-VLAN30-IT
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)# exit
Router(config)#
```

Gambar 11: Konfigurasi Subinterface

6. Konfigurasi Subinterface VLAN 40

A terminal window showing the configuration of a subinterface for VLAN 40. The commands are: 'interface gig0/2.40', 'description GATEWAY-VLAN40-MARKETING', 'encapsulation dot1Q 40', 'ip address 192.168.40.1 255.255.255.0', 'no shutdown', 'exit', and 'Router(config)#'.

```
Router(config)#interface gig0/2.40
Router(config-subif)# description GATEWAY-VLAN40-MARKETING
Router(config-subif)# encapsulation dot1Q 40
Router(config-subif)# ip address 192.168.40.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)# exit
Router(config)#
```

Gambar 12: Konfigurasi Subinterface

7. Konfigurasi DHCP-Server untuk Vlan 10

```

Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10
Router(config)#
Router(config)#ip dhcp pool VLAN10-FINANCE
Router(dhcp-config)# network 192.168.10.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.10.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)#exit

```

Gambar 13: Konfigurasi DHCP-Server

8. Konfigurasi DHCP-Server untuk Vlan 20

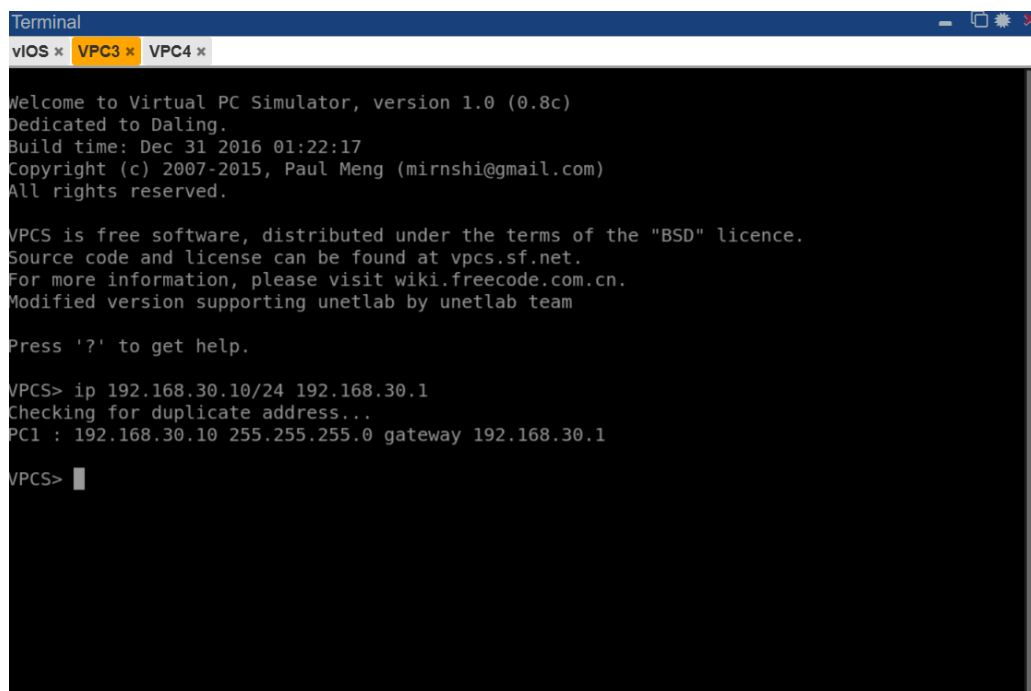
```

Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10
Router(config)#
Router(config)#ip dhcp pool VLAN20-HR
Router(dhcp-config)# network 192.168.20.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.20.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)#exit

```

Gambar 14: Konfigurasi DHCP-Server

9. Konfigurasi IP Static pada VPCS untuk vlan 30 dan 40



```

Terminal
VPCS x VPCS3 x VPCS4 x

Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

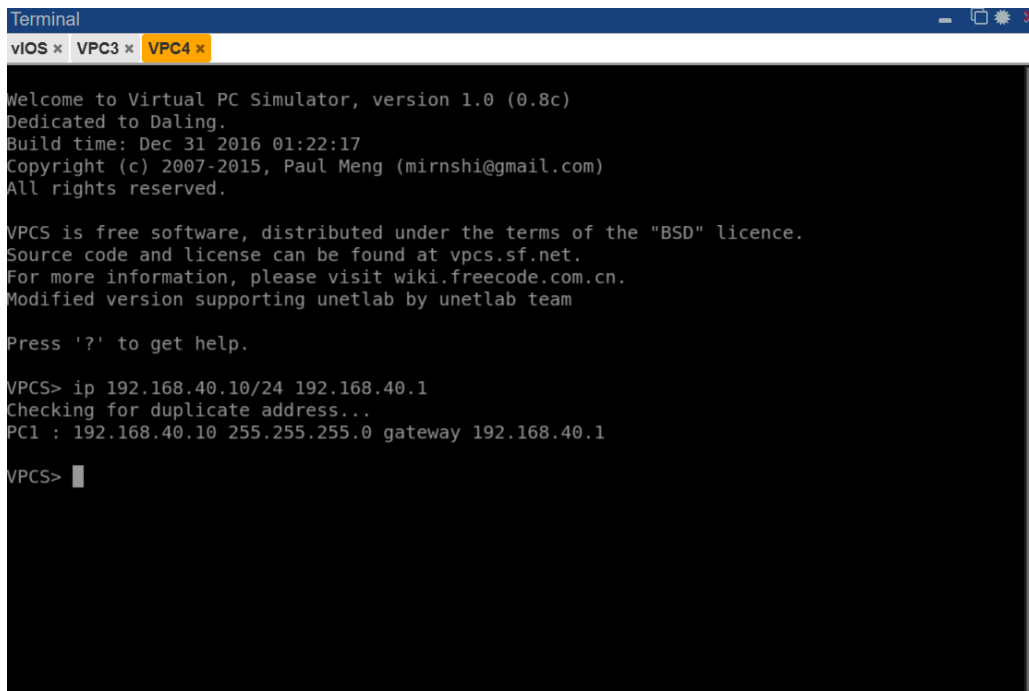
Press '?' to get help.

VPCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
PC1 : 192.168.30.10 255.255.255.0 gateway 192.168.30.1

VPCS>

```

Gambar 15: Konfigurasi IP Static pada vpcs3



```
Terminal
vIOS x VPC3 x VPC4 x

Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
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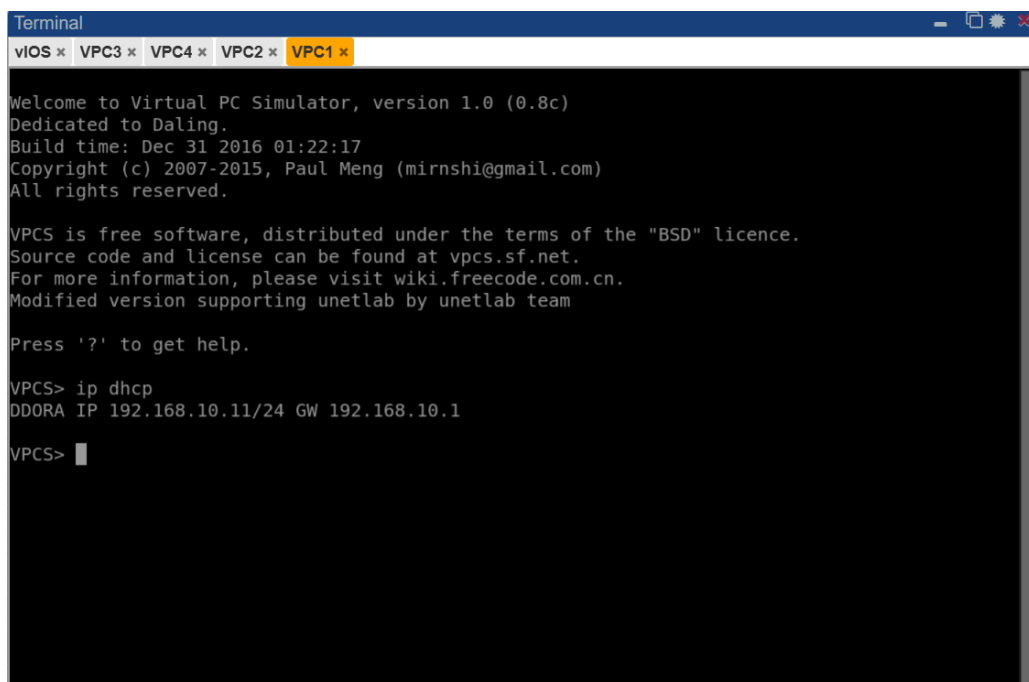
VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> ip 192.168.40.10/24 192.168.40.1
Checking for duplicate address...
PC1 : 192.168.40.10 255.255.255.0 gateway 192.168.40.1

VPCS> 
```

Gambar 16: Konfigurasi IP Static pada vpcs4



```
Terminal
vIOS x VPC3 x VPC4 x VPC2 x VPC1 x

Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
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For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

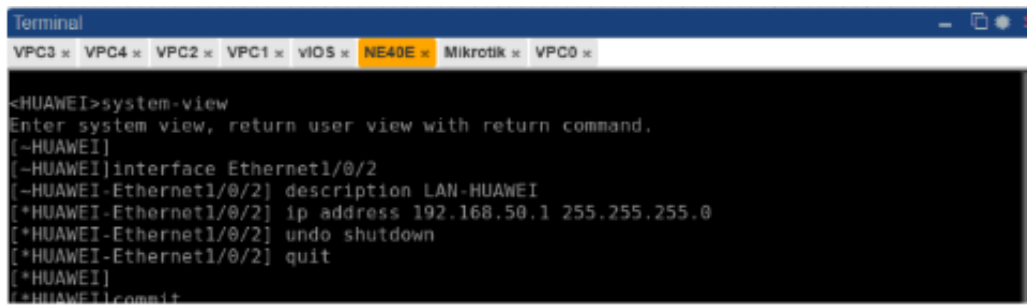
Press '?' to get help.

VPCS> ip dhcp
DDORA IP 192.168.10.11/24 GW 192.168.10.1

VPCS> 
```

Gambar 17: Konfigurasi IP DHCP pada vpcs1 dan vpcs2

10. Verifikasi Cisco Router

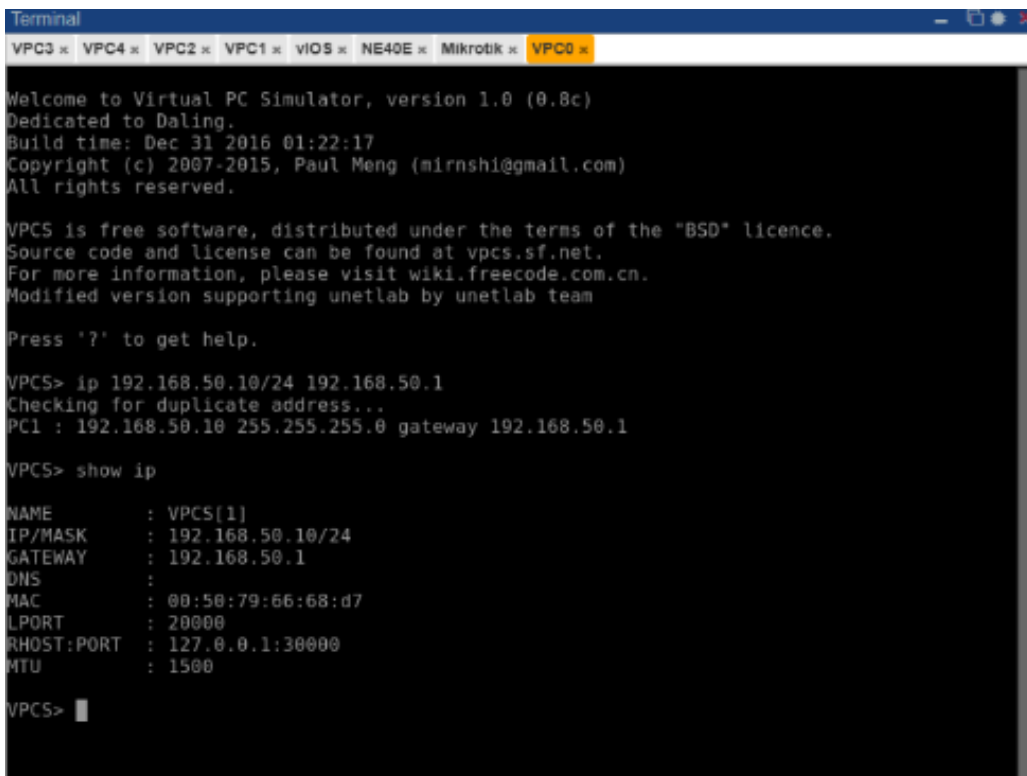


```
Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x

<HUAWEI>system-view
Enter system view, return user view with return command.
[~HUAWEI]
[~HUAWEI]interface Ethernet1/0/2
[~HUAWEI-Ethernet1/0/2] description LAN-HUAWEI
[*HUAWEI-Ethernet1/0/2] ip address 192.168.50.1 255.255.255.0
[*HUAWEI-Ethernet1/0/2] undo shutdown
[*HUAWEI-Ethernet1/0/2] quit
[*HUAWEI]
[*HUAWEI]commit
```

Gambar 22: Konfigurasi Lan Huawei

2. Konfigurasi IP VPC Lan Huawei



```
Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x

Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
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VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.50.10/24
GATEWAY    : 192.168.50.1
DNS        :
MAC        : 00:50:79:66:68:d7
LPORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> █
```

Gambar 23: Set IP VPC Lan

3. Konfigurasi IP link cisco <-> Huawei link 1

```

terminal
VPC3 * VPC4 * VPC2 * VPC1 * VIOS * NE40E * Mikrotik * VPC0 *
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.d6    Jun 07 2026 03:36 AM    Automatic
192.168.20.11    0100.5079.6668.d5    Jun 07 2026 03:36 AM    Automatic
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.d6    Jun 07 2026 03:38 AM    Automatic
192.168.20.11    0100.5079.6668.d5    Jun 07 2026 03:38 AM    Automatic
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#interface gi0/3
Router(config-if)# description LINK1-TO-HUAWEI
Router(config-if)# ip address 10.10.10.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#
*Jun 6 03:43:40.303: %LINK-3-UPDOWN: Interface GigabitEthernet0/3, changed state to up
*Jun 6 03:43:41.309: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/
0, changed state to up
Router(config)#
Router(config)#interface gi0/0
Router(config-if)# description LINK2-TO-HUAWEI
Router(config-if)# ip address 10.10.11.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#

```

Gambar 24: Konfigurasi IP Link cisco ke Huawei

4. Konfigurasi IP Link Cisco <-> Huawei Link 2

```

terminal
VPC3 * VPC4 * VPC2 * VPC1 * VIOS * NE40E * Mikrotik * VPC0 *
[~HUAWEI]interface Ethernet1/0/0
[~HUAWEI-Ethernet1/0/0] description LINK1-TO-CISCO
[*HUAWEI-Ethernet1/0/0] ip address 10.10.10.2 255.255.255.252
[*HUAWEI-Ethernet1/0/0] undo shutdown
[*HUAWEI-Ethernet1/0/0] quit
[*HUAWEI]
[*HUAWEI]commit
[~HUAWEI]
[~HUAWEI]system-view
Error: Unrecognized command found at '^' position.
[~HUAWEI]
[~HUAWEI]interface Ethernet1/0/0
[~HUAWEI-Ethernet1/0/0] description LINK1-TO-CISCO
[~HUAWEI-Ethernet1/0/0] ip address 10.10.10.2 255.255.255.252
Error: The address already exists.
[~HUAWEI-Ethernet1/0/0] undo shutdown
[~HUAWEI-Ethernet1/0/0] quit
[~HUAWEI]
[~HUAWEI]commit
[~HUAWEI]
[~HUAWEI]system-view
Error: Unrecognized command found at '^' position.
[~HUAWEI]
[~HUAWEI]interface Ethernet1/0/4
[~HUAWEI-Ethernet1/0/4] description LINK2-TO-CISCO
[*HUAWEI-Ethernet1/0/4] ip address 10.10.11.2 255.255.255.252
[*HUAWEI-Ethernet1/0/4] undo shutdown
[*HUAWEI-Ethernet1/0/4] quit
[*HUAWEI]
[*HUAWEI]commit
[~HUAWEI]

```

Gambar 25: Konfigurasi IP Link cisco ke Huawei

5. Konfigurasi IP link Huawei <-> Mikrotik

```

[admin@MikroTik] >> /ip address add address=10.10.20.2/30 interface=ether1 comment=T0-HUAWEI
[admin@MikroTik] >> /ip address add address=10.10.30.2/30 interface=ether2 comment=T0-CISCO
[admin@MikroTik] >> ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ::: T0-HUAWEI
10.10.20.2/30 10.10.20.0 ether1
1 ::: T0-CISCO
10.10.30.2/30 10.10.30.0 ether2
[admin@MikroTik] >>

```

Gambar 26: Konfigurasi IP Link Huawei ke Mikrotik

6. Konfigurasi IP link Cisco <-> Mikrotik

```

Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x
*Jun 6 03:44:38.631: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Jun 6 03:44:39.632: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config)#interface gi0/1
Router(config-if)# description LINK-T0-MIKROTIK
Router(config-if)# ip address 10.10.30.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#
*Jun 6 03:46:01.649: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
*Jun 6 03:46:02.649: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config)#
Router(config)#exit
Router#show
*Jun 6 03:46:29.797: %SYS-5-CONFIG I: Configured from console by console
% Type "show ?" for a list of subcommands

```

Gambar 27: Konfigurasi IP Link cisco ke Mikrotik

7. Verifikasi Interface

```

Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x
*Jun 6 03:44:38.631: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Jun 6 03:44:39.632: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config)#interface gi0/1
Router(config-if)# description LINK-T0-MIKROTIK
Router(config-if)# ip address 10.10.30.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#
*Jun 6 03:46:01.649: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
*Jun 6 03:46:02.649: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config)#
Router(config)#exit
Router#show
*Jun 6 03:46:29.797: %SYS-5-CONFIG I: Configured from console by console
% Type "show ?" for a list of subcommands
Router#show ip interface brief
^
% Invalid input detected at '^' marker.

Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.10.11.1 YES manual up up
GigabitEthernet0/1 10.10.30.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset up up
GigabitEthernet0/2.10 192.168.10.1 YES manual up up
GigabitEthernet0/2.20 192.168.20.1 YES manual up up
GigabitEthernet0/2.30 192.168.30.1 YES manual up up
GigabitEthernet0/2.40 192.168.40.1 YES manual up up
GigabitEthernet0/3 10.10.10.1 YES manual up up
Router#display ip interface brief
^

```

Gambar 28: Hasil Verifikasi Interface Cisco Router

```

Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x

down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface                IP Address/Mask      Physical  Protocol  VPN
Ethernet1/0/0            10.10.10.2/30        up        up        --
Ethernet1/0/0.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/1            unassigned           up        down      --
Ethernet1/0/1.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/2            192.168.50.1/24     up        up        --
Ethernet1/0/2.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/3            10.10.20.1/30        up        up        --
Ethernet1/0/3.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/4            10.10.11.2/30       up        up        --
Ethernet1/0/4.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/5            unassigned           up        down      --
Ethernet1/0/5.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/6            unassigned           up        down      --
Ethernet1/0/6.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/7            unassigned           up        down      --
Ethernet1/0/7.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/8            unassigned           up        down      --
Ethernet1/0/8.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/9            unassigned           up        down      --
Ethernet1/0/9.4094       128.1.138.137/16    up        up        l3vpn
GigabitEthernet0/0/0     unassigned           up        down      --
---- More ----

```

Gambar 29: Hasil Verifikasi Interface Huawei Router

8. Verifikasi ping Antar Router

```

Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x

sent=20 received=20 packet-loss=0% min-rtt=0ms avg-rtt=2ms max-rtt=7ms
SEQ HOST                SIZE TTL TIME  STATUS
20 10.10.20.1            56 255 1ms
21 10.10.20.1            56 255 1ms
22 10.10.20.1            56 255 1ms
23 10.10.20.1            56 255 2ms
24 10.10.20.1            56 255 1ms
25 10.10.20.1            56 255 1ms
26 10.10.20.1            56 255 3ms
27 10.10.20.1            56 255 1ms
28 10.10.20.1            56 255 1ms
29 10.10.20.1            56 255 1ms
30 10.10.20.1            56 255 0ms
31 10.10.20.1            56 255 1ms
32 10.10.20.1            56 255 1ms
sent=33 received=33 packet-loss=0% min-rtt=0ms avg-rtt=1ms max-rtt=7ms

[admin@MikroTik] > ping 10.10.20.1
SEQ HOST                SIZE TTL TIME  STATUS
0 10.10.20.1            56 255 2ms
1 10.10.20.1            56 255 2ms
2 10.10.20.1            56 255 1ms
3 10.10.20.1            56 255 2ms
sent=4 received=4 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=2ms

[admin@MikroTik] > ping 10.10.30.1
SEQ HOST                SIZE TTL TIME  STATUS
0 10.10.30.1            56 255 0ms
1 10.10.30.1            56 255 2ms
2 10.10.30.1            56 255 2ms
3 10.10.30.1            56 255 5ms
sent=4 received=4 packet-loss=0% min-rtt=0ms avg-rtt=2ms max-rtt=5ms

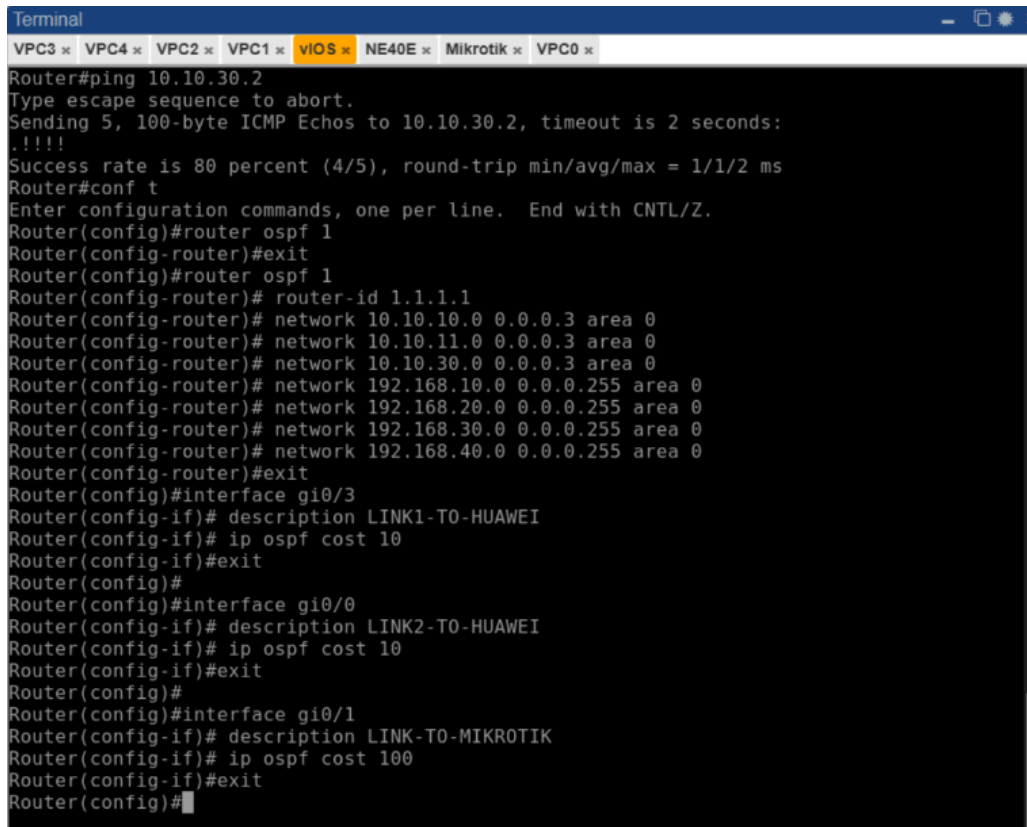
[admin@MikroTik] >

```

Gambar 30: Hasil Verifikasi ping antar router

1.4 Konfigurasi OSPF Multivendor dan OSPF Cost

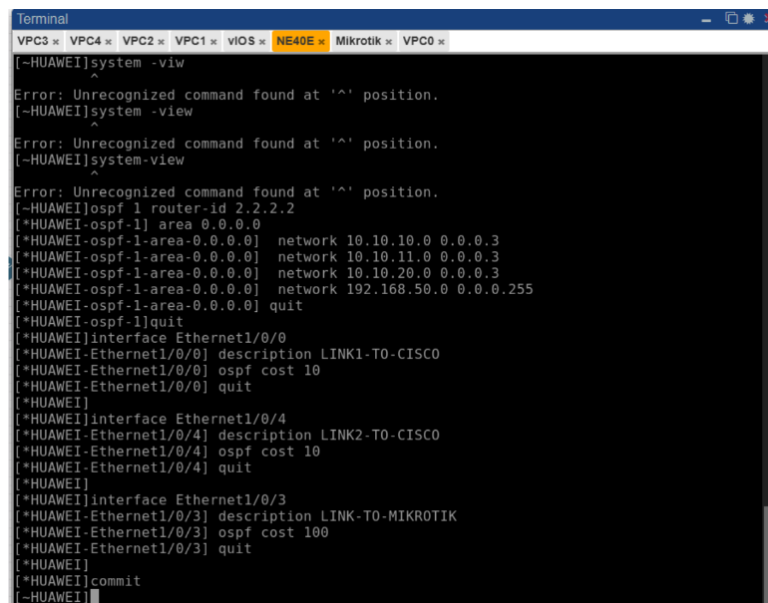
1. Konfigurasi ospf pada Cisco Router



```
Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#exit
Router(config)#router ospf 1
Router(config-router)# router-id 1.1.1.1
Router(config-router)# network 10.10.10.0 0.0.0.3 area 0
Router(config-router)# network 10.10.11.0 0.0.0.3 area 0
Router(config-router)# network 10.10.30.0 0.0.0.3 area 0
Router(config-router)# network 192.168.10.0 0.0.0.255 area 0
Router(config-router)# network 192.168.20.0 0.0.0.255 area 0
Router(config-router)# network 192.168.30.0 0.0.0.255 area 0
Router(config-router)# network 192.168.40.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#interface gi0/3
Router(config-if)# description LINK1-T0-HUAWEI
Router(config-if)# ip ospf cost 10
Router(config-if)#exit
Router(config)#
Router(config)#interface gi0/0
Router(config-if)# description LINK2-T0-HUAWEI
Router(config-if)# ip ospf cost 10
Router(config-if)#exit
Router(config)#
Router(config)#interface gi0/1
Router(config-if)# description LINK-T0-MIKROTIK
Router(config-if)# ip ospf cost 100
Router(config-if)#exit
Router(config)#
```

Gambar 31: Konfigurasi ospf pada Router Cisco

2. Konfigurasi OSPF pada Huawei Router



```
Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x
[~HUAWEI]system -v1w
Error: Unrecognized command found at '^' position.
[~HUAWEI]system -v1w
Error: Unrecognized command found at '^' position.
[~HUAWEI]system-view
Error: Unrecognized command found at '^' position.
[~HUAWEI]ospf 1 router-id 2.2.2.2
[*HUAWEI-ospf-1] area 0.0.0.0
[*HUAWEI-ospf-1-area-0.0.0.0] network 10.10.10.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0] network 10.10.11.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0] network 10.10.20.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0] network 192.168.50.0 0.0.0.255
[*HUAWEI-ospf-1-area-0.0.0.0] quit
[*HUAWEI-ospf-1]quit
[*HUAWEI]interface Ethernet1/0/0
[*HUAWEI-Ethernet1/0/0] description LINK1-T0-CISCO
[*HUAWEI-Ethernet1/0/0] ospf cost 10
[*HUAWEI-Ethernet1/0/0] quit
[*HUAWEI]
[*HUAWEI]interface Ethernet1/0/4
[*HUAWEI-Ethernet1/0/4] description LINK2-T0-CISCO
[*HUAWEI-Ethernet1/0/4] ospf cost 10
[*HUAWEI-Ethernet1/0/4] quit
[*HUAWEI]
[*HUAWEI]interface Ethernet1/0/3
[*HUAWEI-Ethernet1/0/3] description LINK-T0-MIKROTIK
[*HUAWEI-Ethernet1/0/3] ospf cost 100
[*HUAWEI-Ethernet1/0/3] quit
[*HUAWEI]
[*HUAWEI]commit
[~HUAWEI]
```

Gambar 32: Konfigurasi ospf pada Router Huawei

3. Konfigurasi OSPF pada MikroTik

```

Terminal
VPC3 x VPC4 x VPC2 x VPC1 x vIOS x NE40E x Mikrotik x VPC0 x
syntax error (line 1 column 154)
[admin@MikroTik] >> /routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] >> /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] >> routing ospf neighbor print

[admin@MikroTik] > /routing ospf neighbor print

[admin@MikroTik] > /routing ospf interface add network=10.10.20.0/30 area=backbone
syntax error (line 1 column 37)
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone

[admin@MikroTik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=7s

1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=3s
[admin@MikroTik] >

```

Gambar 33: Konfigurasi ospf pada mikrotik

4. Verifikasi OSPF Neighbor

```

Terminal
VPC3 x VPC4 x VPC2 x VPC1 x vIOS x NE40E x Mikrotik x VPC0 x
0 ADo 10.10.10.0/30 10.10.30.1 110
1 ADo 10.10.11.0/30 10.10.20.1 110
2 ADC 10.10.20.0/30 10.10.30.1 110
3 ADC 10.10.30.0/30 10.10.20.1 110
4 ADo 192.168.10.0/24 10.10.30.1 110
5 ADo 192.168.20.0/24 10.10.30.1 110
6 ADo 192.168.30.0/24 10.10.30.1 110
7 ADo 192.168.40.0/24 10.10.30.1 110
8 ADo 192.168.50.0/24 10.10.20.1 110
[admin@MikroTik] >> /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADo 10.10.10.0/30 10.10.30.1 110
1 ADo 10.10.11.0/30 10.10.20.1 110
2 ADo 192.168.10.0/24 10.10.30.1 110
3 ADo 192.168.20.0/24 10.10.30.1 110
4 ADo 192.168.30.0/24 10.10.30.1 110
5 ADo 192.168.40.0/24 10.10.30.1 110
6 ADo 192.168.50.0/24 10.10.20.1 110
[admin@MikroTik] >> /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m51s

1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m47s
[admin@MikroTik] >>

```

Gambar 34: Verifikasi OSPF milik tetangga

5. Verifikasi Routing Table


```

Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x
4 ADo 192.168.30.0/24 10.10.30.1 110
5 ADo 192.168.40.0/24 10.10.30.1 110
6 ADo 192.168.50.0/24 10.10.20.1 110
[admin@MikroTik] >> /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m51s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m47s
[admin@MikroTik] >> routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=8m49s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=8m45s
[admin@MikroTik] >> /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADo 10.10.10.0/30 10.10.30.1 110
1 ADo 10.10.11.0/30 10.10.30.1 110
2 ADo 192.168.10.0/24 10.10.30.1 110
3 ADo 192.168.20.0/24 10.10.30.1 110
4 ADo 192.168.30.0/24 10.10.30.1 110
5 ADo 192.168.40.0/24 10.10.30.1 110
6 ADo 192.168.50.0/24 10.10.20.1 110
[admin@MikroTik] >>

```

Gambar 35: Verifikasi Routing Table

6. Verifikasi Koneksi end-to-end

```

Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x
VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=2.162 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=2.778 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=2.033 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=2.192 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=2.105 ms
VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=3.024 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=3.197 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=1.784 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=2.249 ms
^[[A84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=2.572 ms
VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=3.109 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=2.709 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=3.173 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=2.421 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=2.640 ms
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=9.522 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=4.636 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.330 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=7.060 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.090 ms
VPCS>

```

Gambar 36: Verifikasi koneksi

1.5 Pengujian Failover OSPF

1. Kondisi normal: Dua link cisco-huawei aktif

```
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.20.1    1.863 ms  1.899 ms  1.659 ms
 2  10.10.10.2     3.725 ms  1.951 ms  1.827 ms
 3  *192.168.50.10 2.342 ms (ICMP type:3, code:3, Destination port unreachable)
```

Gambar 37: Hasil Show dua link cisco-huawei aktif

2. Kondisi 1 link cisco-huawei mati

```
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.20.1    2.205 ms  1.889 ms  2.103 ms
 2  10.10.11.2     3.546 ms  1.992 ms  2.673 ms
 3  *192.168.50.10 5.013 ms (ICMP type:3, code:3, Destination port unreachable)
```

Gambar 38: Hasil Show satu link cisco-huawei mati

3. Kondisi dua link cisco-huawei mati

```
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.20.1    2.008 ms  2.293 ms  1.523 ms
 2  10.10.30.2     2.880 ms  2.019 ms  1.656 ms
 3  10.10.20.1     4.773 ms  3.144 ms  3.229 ms
 4  *192.168.50.10 6.816 ms (ICMP type:3, code:3, Destination port unreachable)
```

Gambar 39: Hasil Show dua link cisco-huawei mati

4. Mengaktifkan kembali dua link cisco huawei

```
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.20.1    2.344 ms  6.521 ms  1.836 ms
 2  10.10.10.2     3.347 ms  2.813 ms  2.926 ms
 3  *192.168.50.10 3.669 ms (ICMP type:3, code:3, Destination port unreachable)
```

Gambar 40: Hasil Show pengaktifan dua link cisco-huawei

2 Analisis Hasil Percobaan

Pada praktikum ini seluruh konfigurasi berhasil dilakukan sesuai tujuan yang ditetapkan. Cisco Switch berhasil dikonfigurasi dengan VLAN 10, 20, 30, dan 40 serta trunk port yang mampu membawa seluruh VLAN menuju Cisco Router. Implementasi Router-on-a-Stick pada Cisco Router memungkinkan komunikasi antar-VLAN melalui subinterface yang berfungsi sebagai gateway masing-masing jaringan. Selain itu, DHCP Server berhasil memberikan alamat IP secara otomatis kepada host pada VLAN 10 dan VLAN 20, sedangkan VLAN 30 dan VLAN 40 menggunakan alamat IP statis. Pengujian konektivitas menunjukkan bahwa seluruh host dapat berkomunikasi dengan gateway maupun perangkat pada VLAN lain.

Konfigurasi alamat IP antarrouter pada Cisco Router, Huawei Router, dan MikroTik Router juga berhasil diterapkan. Seluruh router dapat saling melakukan *ping* pada jaringan yang terhubung langsung, menunjukkan bahwa konfigurasi addressing dan konektivitas fisik telah berjalan dengan baik.

Jaringan LAN Huawei pada subnet 192.168.50.0/24 juga berhasil terhubung dengan perangkat lain dalam topologi.

Implementasi OSPF pada ketiga router berhasil membentuk hubungan *neighbor* sehingga setiap router dapat mempelajari rute jaringan lain secara dinamis. Routing table pada Cisco, Huawei, dan MikroTik menunjukkan bahwa seluruh network berhasil dipertukarkan melalui OSPF. Pengujian konektivitas end-to-end membuktikan bahwa seluruh VLAN pada sisi Cisco dapat berkomunikasi dengan LAN Huawei, demikian pula sebaliknya.

Pada pengujian failover, OSPF mampu melakukan konvergensi secara otomatis ketika salah satu maupun kedua link utama Cisco–Huawei dinonaktifkan. Saat kedua link utama gagal, trafik berpindah melalui jalur cadangan yang melewati MikroTik. Setelah link utama diaktifkan kembali, OSPF mengembalikan jalur routing ke rute dengan cost yang lebih rendah sehingga komunikasi tetap berlangsung tanpa perlu konfigurasi ulang secara manual.

3 Hasil Tugas Modul

Hasil [Tugas Modul](#) terdapat pada Github kelompok yang dapat diakses melalui link yang tertera.

4 Kesimpulan

Berdasarkan hasil praktikum yang telah dilakukan, dapat disimpulkan bahwa konfigurasi VLAN, trunking, Router-on-a-Stick, DHCP Server, addressing antarrouter, serta routing dinamis OSPF pada lingkungan multivendor berhasil diterapkan dengan baik. Seluruh perangkat mampu saling berkomunikasi sesuai topologi yang dirancang dan seluruh jaringan berhasil dipertukarkan melalui OSPF. Selain itu, mekanisme failover OSPF terbukti mampu menyediakan redundansi jaringan dengan mengalihkan jalur komunikasi secara otomatis ke link cadangan ketika link utama mengalami gangguan. Dengan demikian, tujuan praktikum untuk memahami implementasi VLAN, routing antarjaringan, OSPF multivendor, dan failover routing telah tercapai dengan baik.

5 Lampiran

5.1 Dokumentasi saat praktikum



Gambar 41